

Project Subject: Crop Production and Management

Project Title: Environmentally-Friendly Food Protection Tablets

Introduction:

Agriculture plays a crucial role in any nation, serving as a primary source of fertilizers and a vital means of livelihood for many people. The production of various grains and cereals is a key component of agriculture. However, protecting these food products from pests and spoilage remains a significant challenge.

Problem:

To safeguard grains such as wheat and corn, sulfur tablets are commonly used. Despite their effectiveness, sulfur tablets pose serious risks; they are toxic to both humans and animals. Accidental ingestion of these tablets can be fatal. Furthermore, when these chemical tablets are discarded into the environment, they contribute to environmental pollution.

Solution:

In response to these concerns, I propose the development of environmentally-friendly food protection tablets using commonly available household materials. This approach offers a sustainable and eco-friendly alternative for protecting food items from pests.

Required Materials:

- 25 grams cloves
- 25 grams cinnamon
- 10 grams bay leaves
- 50 grams garlic
- 50 grams neem leaves
- 35 grams ash from burnt coconut shells
- Water
- A large bowl or plate
- Mixer
- Other miscellaneous items

Construction:

1. Place all the listed ingredients into a mixer.
2. Add half a cup of water to the mixture.
3. Grind the mixture thoroughly for approximately three minutes to achieve a fine, homogeneous blend.
4. Transfer the mixture to a large plate or bowl.
5. Incorporate the ash from burnt coconut shells into the mixture, ensuring it is well-mixed.
6. Shape the mixture into small tablets.
7. Dry the tablets under sunlight until they are completely dry.

8. Once dried, wrap the tablets in small cloth pieces to create pouches.
9. These prepared food protection tablets can be placed within containers of grains, pulses, or spices to protect them from pests and spoilage.

Working Principle:

The food protection tablets, when placed in storage containers with grains, pulses, or spices, help to prevent pest infestation and spoilage. The natural ingredients in these tablets act as deterrents to pests, safeguarding the stored food without the need for harmful chemicals.

Cost-effectiveness:

The production of these food protection tablets is highly cost-effective. Since the materials used are readily available at home or in the local environment, the overall expense is minimal.

Usage:

These tablets are designed to be used in the storage of grains, pulses, and spices to prevent damage and contamination by pests.

Conclusion:

Traditional chemical pesticides used for food protection are not environmentally friendly and can pose significant health risks. The proposed homemade food protection tablets offer a safe, sustainable, and cost-effective solution for preserving food items. This approach not only ensures effective pest control but also aligns with environmental conservation principles.